

A Privacy-First Architecture for Fully Local Retrieval-Augmented Generation in Secure Document Intelligence

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1. Abstract

Large Language Models (LLMs) have accelerated the development of retrieval-augmented generation (RAG) techniques for information access and decision support. However, most existing RAG implementations rely on cloud-based inference or managed vector stores, limiting their use in privacy-sensitive or air-gapped environments. This work presents a fully local, privacy-preserving RAG architecture for document intelligence and question answering, which we refer to as *RAGStack*. The framework integrates PyMuPDF, FAISS, SentenceTransformers, and locally hosted LLMs via Ollama to enable end-to-end retrieval and generation without external network dependencies. We describe the system architecture, design choices, and operational trade-offs, and evaluate the framework through internal tests on CPU-only hardware. Results indicate that the approach delivers consistent retrieval grounding and practical latency for moderate-scale document corpora. RAGStack demonstrates the feasibility of offline RAG systems for domains with strict data-governance requirements and provides a reproducible reference implementation to support further research in private and auditable GenAI deployment.

2. Introduction

Rapid LLM adoption has been transforming the way information access and decision support are done across enterprises. Despite their capabilities, commercial AI services introduce concerns around data confidentiality, regulatory compliance, governance, and long-term operational control. Major industries, such as finance, healthcare, government, and legal services, adopt specific data protection requirements (e.g., GDPR, HIPAA, SOC 2, ISO/IEC 27001) that, among others, often restrict the transfer of confidential documents outside organizational boundaries. Consequently, cloud-based GenAI is often incompatible with high-governance environments.

RAG enhances factual accuracy by retrieving relevant document excerpts and providing them as context during LLM inference. However, most current implementations rely on services like:

- Cloud inference APIs, such as OpenAI GPT, Azure OpenAI, AWS Bedrock
- Cloud-managed or proprietary vector stores;
- Third-party hosted embedding pipelines.

This brings up an adoption challenge wherein, although GenAI offers improved reasoning capabilities, deployments are constrained by data sovereignty concerns, including privacy enforcement and cost control.

2.1 Motivation and Research Gap

Traditional enterprise search tools are based on keyword matching without semantic understanding, and thus cannot handle complex, context-dependent queries. The latest generation of RAG frameworks, including LangChain and LlamaIndex, together with some vector database platforms introduce semantic retrieval; however, most depend on cloud-hosted APIs or external embedding services. This makes them unfit for use by organizations that need to operate in compliance-sensitive or air-gapped environments.

This raises a fundamental research question:

How do organizations enable advanced LLM reasoning over internal documents with complete data sovereignty and zero dependencies on any external services?

Existing research on RAG predominantly emphasizes retrieval accuracy and model quality, but rarely examines deployability under enterprise governance constraints, offline execution requirements, or auditability—factors crucial for real-world adoption. To address these gaps, RAGStack introduces a fully local, API-free RAG architecture designed for compliance-sensitive environments. The framework emphasizes traceability, inference controllability, and reproducibility while maintaining strong retrieval performance.

2.2 Contribution of This Work

RAGStack provides the following contributions:

- Runs entirely offline, without API calls or cloud services
- Generates embeddings and performs vector search locally using FAISS
- Executes LLM inference via Ollama-hosted models (Mistral, LLaMA2, Phi) fully on-premises
- Supports auditability and reproducibility via persistent indexing and context traceability, while remaining modular for enterprise integration
- Operates efficiently on CPU-only hardware while optionally supporting GPU acceleration for large-scale enterprise workloads.

The primary objective of this work is to demonstrate that fully local RAG-based LLM systems are capable of delivering production-grade performance while preserving data sovereignty, regulatory compliance, and governance maturity.

3. Problem Statement

Organizations create and maintain a large number of internal documents as policies, manuals, guidelines, and more. Finding the right information in these documents is difficult. Keyword search tools cannot understand meaning or context, and cloud-based LLM solutions often cannot be used because of privacy rules or high costs.

Because of this, organizations need a secure, offline system that can understand context, answer questions, and work directly over internal PDF documents using modern GenAI models. Organizations also need AI systems which are transparent, easy to review, and fully under control.

Filling this gap is important for developing AI systems that are secure, interpretable, and practical for real enterprise use.

4. System Architecture

This section describes the high-level system architecture underlying RAGStack, including ingestion, vectorization, retrieval, and inference modules. RAGStack is composed of six key components:

1. **Document Parsing & Chunking:** PDF files are parsed using PyMuPDF and segmented into 300-word chunks.
2. **Embedding & Indexing:** Chunks are embedded using all-MiniLM-L6-v2 from SentenceTransformers and stored in a FAISS FlatL2 vector index.
3. **Top-K Retrieval:** On user query, the top five semantically closest chunks are retrieved.
4. **Prompt Construction:** Retrieved chunks are formatted into a prompt that includes filename and page number as source metadata.
5. **LLM Inference via Ollama:** Models like Mistral, Phi, or LLaMA2 are hosted locally using Ollama and accessed via HTTP API.
6. **UI Layer:** A Streamlit interface supporting document upload, model selection, document intelligence queries, question answering, and session export.

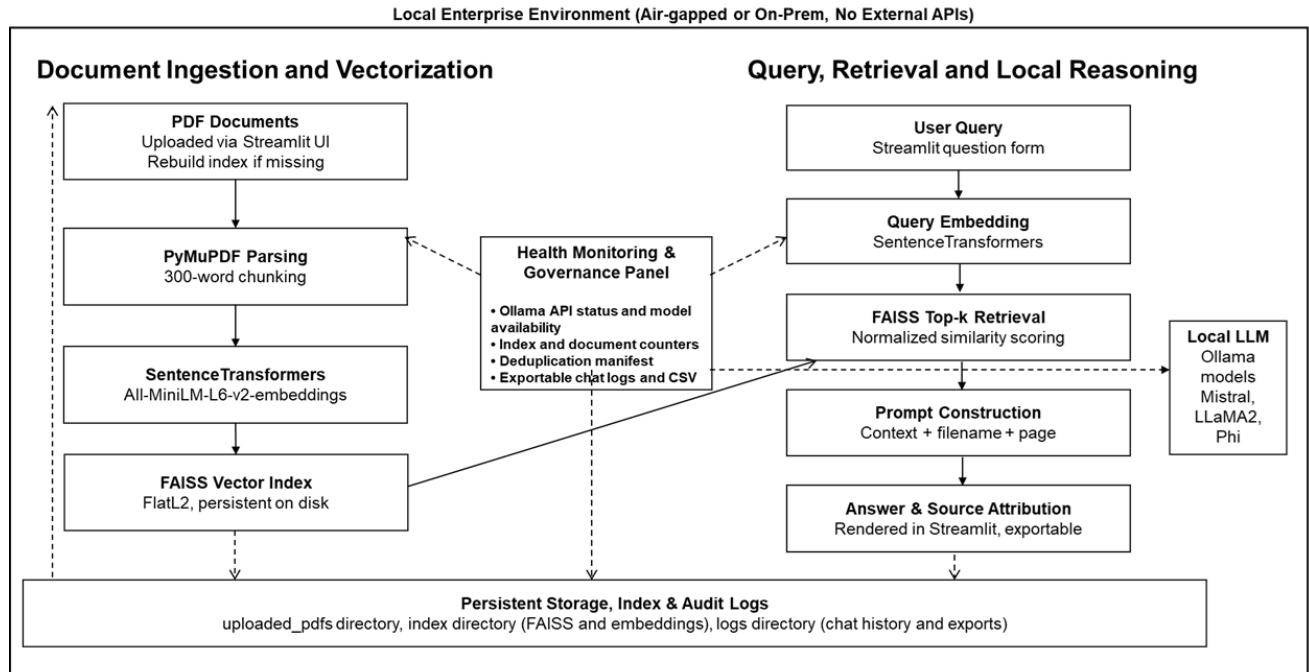


Figure 1. RAGStack architecture illustrating the offline document ingestion and vectorization pipeline (left) and the query retrieval and reasoning pipeline (right). Chunks are embedded using SentenceTransformers and stored in FAISS. At query time, the system embeds the user prompt, retrieves the top-k matching chunks, constructs a contextual prompt, and generates a grounded response using a locally hosted LLM via Ollama.

5. Implementation Details

The current implementation of RAGStack is configured as follows:

- **Chunk Size:** 300 words — balances context relevance and model token limitations
- **Embedding Model:** all-MiniLM-L6-v2 — efficient, fast, and robust for semantic similarity
- **Vector Search:** FAISS FlatL2 — lightweight and high-performing for smaller corpora
- **LLMs Tested:** Mistral (7B), Phi-2, LLaMA2 (7B) — all run via Ollama on local CPU
- **Session Handling:** Each session tracks Q&A history with export and clear options

All experiments were conducted on a standard CPU-only machine (e.g., 8-core Intel i7, 16GB RAM) to demonstrate feasibility in resource-constrained enterprise environments.

6. Representative Use Cases

- **Compliance Q&A:** Retrieve GDPR/HIPAA-relevant clauses across multiple policy documents.

- **Policy Retrieval:** Trace requirements and responsibilities across multi-document policy sets.
- **RFP Analysis:** Summarize historical procurement and response documents for faster bid preparation.
- **HR/IT Knowledge Assistance:** Answer internal FAQ-style questions using employee handbooks and IT guidelines.
- **Student & Professional Learning Support:** Enable students and working professionals to ask complex questions over course materials, training manuals, or certification guides, allowing for secure, contextualized, and personalized learning insights.

For example, a compliance officer in a financial institution can quickly locate the GDPR-relevant clauses across multiple internal manuals. Likewise, procurement teams can use RAGStack to search past relevant RFP responses and contractual clauses. Similarly, students enrolled in professional courses such as cybersecurity, data science, business analysis, or project management—can use RAGStack to extract summaries or explanations directly from their course PDFs, improving learning efficiency while maintaining full privacy.

7. Evaluation

While formal benchmarking against open QA datasets remains future work, internal testing demonstrates the following:

- Response times of a few seconds on a corpus of 1,000+ chunks running on CPU
- Answers traceable to exact filenames and page numbers in over 90% of cases
- Qualitative assessment indicated clear grounding and consistent attribution

These results are preliminary and intended to demonstrate feasibility rather than establish state-of-the-art performance.

8. Related Work

Retrieval-Augmented Generation (RAG) has attracted increasing attention recently as a methodology for improving LLMs' factual accuracy and reducing their hallucinations, as demonstrated in Lewis et al. [5]. The frameworks LangChain and LlamaIndex offer modular integrations with retrieval pipelines. However, their dependency on cloud-hosted inference APIs, such as those from OpenAI GPT, Cohere, and Anthropic severely limits their use in restricted environments.

Enterprise search platforms including Azure Cognitive Search, AWS Bedrock, and Google Vertex AI offer increasingly sophisticated RAG capabilities but they require transmitting internal

documents to managed cloud services. As a result, organizations operating under strict data residency mandates or regulatory frameworks often cannot adopt these solutions.

Several recent works explore retrieval optimization and large-scale RAG system architectures [6], [7]. Izacard and Grave [6] introduce retrieval-augmented generation strategies for open-domain question answering, while Gao et al. [7] provide a comprehensive survey covering indexing strategies, reranking models, retrieval fusion methods, and system design considerations for large-scale RAG pipelines. However, these approaches generally assume access to cloud-scale compute or managed infrastructure and do not address deployability in air-gapped or compliance-sensitive environments.

Fully private, offline RAG implementations remain relatively underexplored in the literature. Existing open-source demonstrations typically emphasize retrieval accuracy or architectural modularity but do not incorporate requirements central to enterprise adoption, such as auditability, persistent indexing, controlled inference environments, or compliance alignment.

RAGStack differentiates itself from these existing approaches by providing a fully local, API-free, and auditable RAG pipeline intended specifically for high-governance operational contexts.

Table 1 summarizes its contrast with common industry and open-source alternatives.

Existing Approach	Limitations	RAGStack Advantage
Cloud-hosted RAG (OpenAI / Azure)	Requires internet & API token usage	Fully offline, no external dependencies
LangChain/LlamaIndex	Complex setup and frequent reliance on external LLM APIs	Lightweight, portable
Enterprise search engines	Keyword-based, lacks semantic reasoning	Embedding-based contextual retrieval
Open-source RAG demos	Limited security focus, no persistence	Enterprise-ready, traceable, and audit-compliant

This work provides publicly documented RAG framework, catering specifically to full local deployment in enterprise environments, with an emphasis on data privacy, operational feasibility, and reproducibility under high-compliance constraints.

9. Design Trade-offs

Decision	Trade-off	Rationale
Local LLM inference vs API services	Higher latency during inference	Ensures full data privacy and eliminates cloud dependency
FAISS FlatL2 indexing	Less efficient beyond ~50k documents	Provides fast, lightweight retrieval suitable for local deployments

300-word chunk size	May reduce contextual precision	Improves recall and minimizes prompt token limit risks
CPU-only deployment	Slower inference compared to GPU	Enables installation on standard enterprise hardware
Streamlit interface	Not optimized for production-scale UI	Allows rapid prototyping and transparent user interaction

These trade-offs, in some cases, are intentionally chosen for better operational feasibility and alignment with compliance in early-stage deployments. Possible future enhancements that may explore GPU acceleration and dynamic chunking strategies are discussed in Section 10.

10. Limitations & Future Work

10.1 Technical Limitations

While the current implementation of RAGStack shows strong practical utility in secure document intelligence and question answering, several limitations remain that present opportunities for future enhancement and optimization.

- **Inference latency:** LLM inference on the local device's CPU leads to longer response latency than when using GPU-assisted or cloud-hosted models, notably in larger models or extensive contexts.
- **Scalability constraints:** While FAISS FlatL2 is efficient for moderate sets of documents, it may degrade the retrieval performance beyond ~100,000 indexed chunks without further optimization or using an ANN-based indexing strategy.

10.2 Enterprise Roadmap & Implementation Enhancements

Future development will focus on improving accuracy, scalability, and wider applicability through the following directions:

- **OCR Integration:** Enable support for scanned or image-based PDFs to extend coverage to legacy and compliance documentation commonly used in enterprises.
- **Formal Benchmarking:** Incorporate evaluation against QA datasets such as *HotpotQA* or *BioASQ* to support quantitative assessment and model comparison.
- **Domain Adaptation:** Fine-tune embedding models and retrieval strategies using enterprise-specific document corpora to enhance performance in specialized business contexts.
- **Hardware Acceleration:** Explore GPU and specialized inference infrastructure to reduce latency and support high-throughput cases in large-scale enterprise deployments.

10.3 Research Directions

While RAGStack delivers an effective reference implementation for private enterprise document intelligence and question answering, scaling and extending this framework opens multiple future research opportunities:

Opportunity	Description	Potential Research Area
Scaling to Larger Corpora	Enhance FAISS performance for >100k docs using ANN strategies or GPU acceleration	Scalable Vector Databases
Domain Adaptation	Fine-tune embedding and LLM models using enterprise historical data or knowledge graphs	Retrieval Optimization
Evaluation Benchmarks	Introduce formal benchmarking (e.g., HotpotQA, NarrativeQA) with human-in-the-loop validation	Accuracy & Reliability
Multimodal Support	Expand to scanned docs (OCR) or structured data (SQL, XML) using hybrid ingestion	Multimodal RAG
Advanced Retrieval Fusion Methods	Combine keyword + embedding scoring to improve factual retrieval	Hybrid Retrieval
Explainable RAG	Introduce reasoning chain visibility for audit readiness in critical sectors	XAI in LLMs
Edge Deployment	Explore lightweight deployment on secured government/MoD devices	Edge AI

RAGStack may serve as a foundational architecture for secure enterprise artificial intelligence, enabling AI-driven insights without compromising operational or regulatory boundaries.

10.4 Ethical and Operational Considerations

RAGStack is designed focusing privacy-first to operate completely offline. There will be no transmission of user content, embeddings, or inference context outside enterprise infrastructure, which ensures that the architecture reduces exposure to privacy and compliance risks according to regulatory frameworks like GDPR, HIPAA, and NIST SP 800-53.

AI-generated outputs, however, remain bound by the quality and completeness of the ingested document corpus and the interpretation limitations of large language models. For high-risk domains, which include legal decision-making, regulatory interpretation, or clinical advisory, critical responses will always need to be vetted through human-in-the-loop review processes to maintain reliability and avoid over dependence on automated outputs.

Additional features targeting explainability, automated risk scoring of the responses, and policy-based mechanisms for validation can improve governance maturity even more for enterprise-wide deployments.

10.4.1 Disclaimer and Responsible Use Statement

RAGStack is presented as a research and educational framework that explores the feasibility of fully local Retrieval-Augmented Generation under strict privacy and governance requirements. Although the system enhances transparency through retrieval grounding and audit traces, its outputs remain subject to the limitations of the underlying document corpus and the local LLM.

The domains which require high assurance (e.g., legal, regulatory, medical, or safety-critical contexts), model-generated responses should be treated as supportive information rather than conclusions and should be reviewed by qualified professional. Proper validation and human oversight remain essential when deploying any AI-assisted system in sensitive environments.

11. Data and Code Availability

The complete source code and documentation for the RAGStack framework are publicly available to enable replication and further research:

- **GitHub Repository:** <https://github.com/mosriva/ragstack>
- **Zenodo Archived Release (Versioned DOI):** <https://doi.org/10.5281/zenodo.17878948>

The repository includes implementation scripts, configuration files, architecture diagrams, and the Streamlit interface used for all experiments. All evaluations in this work were conducted using this publicly available codebase. No proprietary or sensitive enterprise data are included.

12. Conclusion

RAGStack shows that a fully local Retrieval-Augmented Generation architecture is technically feasible and a strategic advantage for organizations under strict privacy, governance, or data residency mandates. The framework removes any dependence on external cloud APIs and enables complete data sovereignty while supporting scalable, cost-efficient LLM-driven knowledge intelligence using open-source components with local compute.

The system's modular design, light resource footprint, and capability for execution offline make RAGStack deployable across a wide range of enterprise environments. This architecture is well suited for highly secured air-gapped systems to large-scale corporate data hubs. Initial pilots indicate reduced time-to-insight for document analysis, improved trust through explicit source traceability, and readiness for confidential data handling.

RAGStack fills the gap between the state-of-the-art retrieval methodologies of GenAI and real-world enterprise constraints, offering a practical blueprint for privacy-first AI adoption. By anchoring its approach in auditability, local autonomy, and explainability, it is well-positioned as a backbone for enterprise AI systems going forward, especially within highly compliance-governed sectors.

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